The Product Topology

Defn: Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. Let all sets of the form:

$$\prod_{\alpha \in J} U_\alpha$$

form a basis for the product space $\prod_{\alpha \in J} X_\alpha$, where U_α is open in X_α $\forall \alpha \in J$. The topology generated by this basis is called the box topology.

Note, $\prod_{\alpha \in J} X_\alpha \ni x$ $\forall x \in \prod_{\alpha \in J} X_\alpha$, and $\prod_{\alpha \in J} U_\alpha \cap \prod_{\alpha \in J} V_\alpha$ is equivalent to $\prod_{\alpha \in J} (U_\alpha \cap V_\alpha)$, so this is indeed a basis.

We can also use the subbasis definition.

Let $\pi_\beta : \prod_{\alpha \in J} X_\alpha \to X_\beta$ be defined by $\pi_\beta : (x_\alpha)_{\alpha \in J} \mapsto x_\beta$.

This is called the projection mapping associated with β.

Note, $(x_\alpha)_{\alpha \in J}$ is just an arbitrary element of $\prod_{\alpha \in J} X_\alpha$.

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Defn: Let $S_\beta := \{\pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ open in } X_\beta\}$ and let

$$S := \bigcup_{\beta \in J} S_\beta.$$

The topology generated by the subbasis S is called the product topology, and under this topology, $\prod_{\alpha \in J} X_\alpha$ is called the product space.

Let $\beta_1, \ldots, \beta_n$ be a finite set of distinct indices in J, and let U_{β_i} be open in X_{β_i}, $i = 1, \ldots, n$. Then

$$B = \pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \pi_{\beta_2}^{-1}(U_{\beta_2}) \cap \cdots \cap \pi_{\beta_n}^{-1}(U_{\beta_n})$$

is an element of $\mathcal{B}$, the basis generated by the product topology.

This basis can be described differently as follows:

$x \in \prod_{\alpha \in J} X_\alpha$ is in B iff its β_ith coordinate is in

$x \in \prod_{\alpha \in J} X_\alpha$ is in B iff its β_1^{th} coordinate is in U_{β_1}, ... its β_n^{th} coordinate is in U_{β_n}. If α is not an index in $\beta_1, \ldots, \beta_n$, then X_α can be anything. So,

$$B = \prod_{\alpha \in J} U_\alpha,$$

where $U_\alpha = X_\alpha$ if $\alpha \neq \beta_1, \ldots, \beta_n$.

Theorem: (Comparison of Box and Product topologies).

The Box topology on $\prod X_\alpha$ has as a basis all sets of the form $\prod U_\alpha$, where U_α open in X_α for each α. The product topology on $\prod X_\alpha$ has as a basis all sets of the form $\prod U_\alpha$ where U_α is open in X_α, and $U_\alpha = X_\alpha$ for all but finitely many α.